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328713 (28)

BE (7th Semester)

Examination, April-May, 2013

Branch : EEE, Et & T

MICROCONTROLLER & EMBEDDED SYSTEM

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Assume suitable data if required. Part (a) is compulsory, attempt any two parts from b, c & d.

Q. 1. (a) In which year 8051 microcontroller was introduced ? 2

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(b) Discuss the speed and packaging of 8051 microcontroller from ATMEL. 7

(c) What is the occupancy of Port 0 on 8051 μc ? Why external register of 10 kΩ pull up register is connected to each pin of Port 0 ? 7

(d) Explain OTP version of 8051 microcontroller and UV-EPROM version of microcontroller. 7

Q. 2. (a) What is machine cycle of 8051 ? 2

(b) Explain the procedure to generate hex file from asm file. 7

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(c) Write a program to transfer value 41H serially (one bit at a time) via pin P2.1. Put two highs at the start and end of the data. Send the byte LSB first. 7

(d) Assume that bit 2.2 is used to control an outdoor light and bit P2.5 a light inside a building. Show how to turn on the outside light and turn off the inside one. 7

Q. 3. (a) Write an instruction to monitor the TF bit. 2

(b) Write an assembly language program to generate following waveform on P1.5. 7

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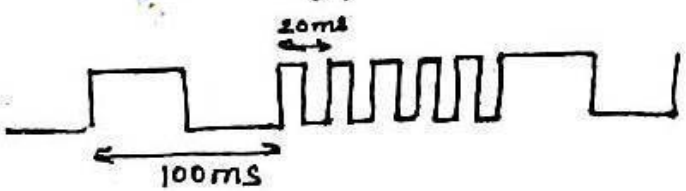

(c) Write an assembly language program to design a counter system to monitor the attendance of a cricket stadium restricted to 66,000. Display the attendance on P1 and P2 in binary. Also draw the interfacing diagram. Use timer 0 and XTAL = 22 MHz. 7

(d) The 8051 microcontroller is to be used as a frequency divider. Write an ALP for the same. Use pin P1.2 and XTAL = 22 MHz. 7

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Q. 4. (a) Sending data to a printer is _____ communication. 2

(b) Explain how baudrate is generated using timer 1 mode 2 with suitable example. 7

(c) Write a program to transfer a letter "V" serially at 19200 baud continuously and also send a letter "M" through port 0, which is connected to display device. Draw the schematic for the same. 7

(d) Write a program to generate a square wave at P1.2 using timer 0 in mode 1, interrupt

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mode, take data from port P2 and send it serially and continuously, when INTO is activated, port P0 is mode 0 for a short time, to switch off the LEDs connected to it. The LEDs will also remain off if the switch connected to INTO pin (P3.2) is kept pressed. 7

Q. 5. (a) In reading the columns of a keyboard matrix, if no key is pressed we should get all _____. 2

(b) Two switches are connected to pins P0.1 and P0.2. They are also vectored to interrupt

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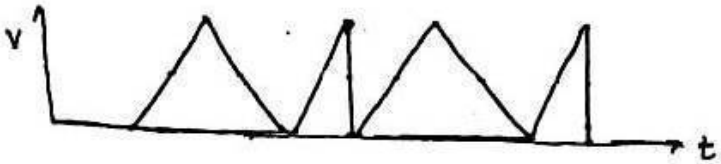
location 0003H, i.e. INTO. Write a program to test which key is pressed, or to verify if both keys are pressed. 7

(c) Draw the interfacing diagram of 0808 ADC with 8051 microcontroller, also explain the programming steps along with suitable wave-forms. 7

(d) Interface DAC with 8051 microcontroller to generate following waveform. 7



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